Find a Gene Assignment Part 1

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> Q1. Tell me the name of a protein you are interested in. Include the species, accession number and known function. This can be a human protein or a protein from any other species as long as it's function is known.
> If you do not have a favorite protein, select human RBP4 or KIF11. Do not use beta globin as this is in the worked example report that I provide you with online.

Name: Interleukin-36 gamma (IL36G)

Species: Homo sapiens *Accession:** Q9NZH8

Function: Interleukin-36 gamma (IL36G) plays a role in skin inflammation and issume signaling. IL36G is associated in skin disorders such as psoriasis and dandruff.

> Q2. Perform a BLAST search against a DNA database, such as a database consisting of genomic DNA or ESTs. The BLAST server can be at NCBI or elsewhere. Include details of the BLAST method used, database searched and any limits applied (e.g. Organism).

Method: TBLASTN search against reptilia nt

Database: Nucleotide collection (nt)

Organsim: Reptilia (taxid:8459)

> Also include the output of that BLAST search in your document. If appropriate, change the font to Courierdd size 10 so that the results are displayed neatly. You can also screen capture a BLAST output (e.g. alt print screen on a PC or on a MAC press -shift-4. The pointer becomes a bulls eye. Select the area you wish to capture and release. The image is saved as a file called Screen Shot [].png in your Desktop directory). It is not necessary to print out all of the blast results if there are many pages.

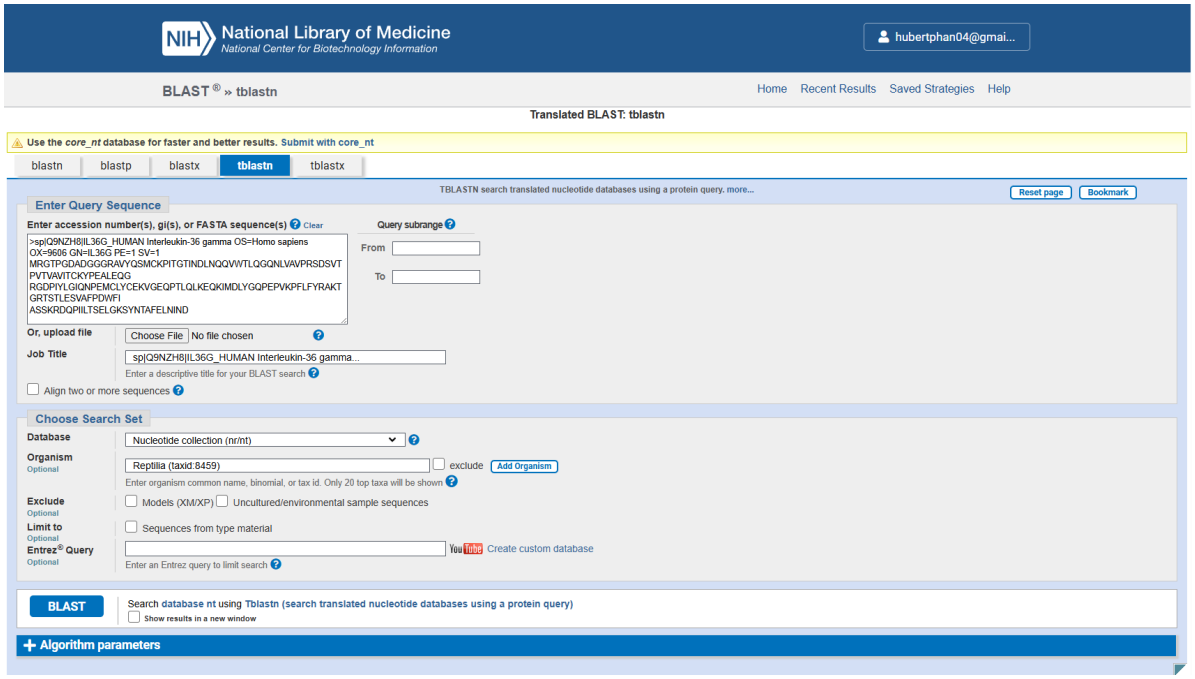

Figure 1: BLAST QUERY

On the BLAST results, clearly indicate a match that represents a protein sequence, encoded from some DNA sequence, that is homologous to your query protein. I need to be able to inspect the pairwise alignment you have selected, including the E value and score. It should be labeled a "genomic clone" or "mRNA sequence", etc. - but include no functional annotation.

Chosen Match: Accession OZ346534.1, Trionyx triunguis genome assembly. See alignment details below.

Edit Search | Save Search | Search Summary | How to read this report? | BLAST Help Videos | Back to Traditional Results Page

Your search is limited to records **include: Reptilia (taxid:8459)**

Job Title	**sp|Q9NZH8|IL36G_HUMAN Interleukin-36 gamma...**
RID	RU2HMSBS016 *Search expires on 02-01 02:44 am* Download All
Program	TBLASTN Citation
Database	nt See details
Query ID	lcl|Query_6676515
Description	sp|Q9NZH8|IL36G_HUMAN Interleukin-36 gamma OS=Hor ...
Molecule type	amino acid
Query Length	169
Other reports	

Filter Results

Organism *only top 20 will appear* exclude

Type common name, binomial, taxid or group name

+ Add organism

Percent Identity to **E value** to **Query Coverage** to

Filter **Reset**

Descriptions | Graphic Summary | Alignments | Taxonomy

Sequences producing significant alignments Download Select columns Show 100

select all *73 sequences selected* GenBank Graphics

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
PREDICTED: Chelonoidis abingdonii interleukin-36 receptor antagonist protein-like (LOC116819404), m...	Chelonoidis abi...	95.5	95.5	90%	1e-20	35.29%	3011	XM_032771100.2
PREDICTED: Emys orbicularis interleukin-36 receptor antagonist protein-like (LOC135873734), mRNA	Emys orbicularis	87.8	87.8	89%	1e-19	35.53%	576	XM_065398370.1
PREDICTED: Chelonoidis abingdonii interleukin-36 receptor antagonist protein-like (LOC116819405), m...	Chelonoidis abi...	87.8	87.8	89%	1e-19	34.87%	576	XM_032771101.1
PREDICTED: Gopherus flavomarginatus interleukin-36 receptor antagonist protein-like (LOC127044660)...	Gopherus flavo...	87.4	87.4	89%	2e-19	34.87%	576	XM_050939738.1
PREDICTED: Malaclemys terrapin pileata interleukin-36 receptor antagonist protein-like (LOC12883275...	Malaclemys terr...	90.1	90.1	85%	2e-19	36.81%	1090	XM_054020339.1
PREDICTED: Mauremys reevesii interleukin-36 receptor antagonist protein-like (LOC120398387), transc...	Mauremys reev...	88.2	88.2	84%	3e-19	37.06%	764	XM_039525790.1
PREDICTED: Mauremys mutica interleukin-36 receptor antagonist protein-like (LOC123364511), transcri...	Mauremys mutica	90.9	90.9	90%	3e-19	35.95%	2253	XM_045006715.1
PREDICTED: Mauremys mutica interleukin-36 receptor antagonist protein-like (LOC123364511), transcri...	Mauremys mutica	90.9	90.9	90%	3e-19	35.95%	2267	XM_045006714.1
PREDICTED: Mauremys mutica interleukin-36 receptor antagonist protein-like (LOC123364511), transcri...	Mauremys mutica	90.9	90.9	90%	3e-19	35.95%	2207	XM_045006713.1
PREDICTED: Chrysemys picta bellii interleukin-36 receptor antagonist protein (LOC101949597), mRNA	Chrysemys pict...	90.1	180	85%	4e-19	36.81%	1357	XM_005279282.4
PREDICTED: Trachemys scripta elegans interleukin-36 receptor antagonist protein-like (LOC117871587...	Trachemys scri...	90.1	90.1	85%	5e-19	36.81%	1389	XM_034759399.1
PREDICTED: Mauremys reevesii interleukin-36 receptor antagonist protein-like (LOC120398387), transc...	Mauremys reev...	88.2	88.2	84%	5e-19	37.06%	836	XM_039525788.1
PREDICTED: Emydura macquarii macquarii interleukin-36 receptor antagonist protein-like (LOC1371479...	Emydura macq...	87.8	87.8	96%	6e-19	32.75%	821	XM_067540356.1
PREDICTED: Mauremys reevesii interleukin-36 receptor antagonist protein-like (LOC120398387), transc...	Mauremys reev...	88.2	88.2	84%	6e-19	37.06%	896	XM_039525787.1
PREDICTED: Carettochelys insculpta interleukin-36 receptor antagonist protein-like (LOC142004348), m...	Carettochelys in...	85.5	85.5	91%	7e-19	34.19%	529	XM_074981942.1
PREDICTED: Mauremys mutica interleukin-36 receptor antagonist protein-like (LOC123365076), transcri...	Mauremys mutica	88.2	88.2	82%	1e-18	36.69%	1072	XM_045007705.1
PREDICTED: Emydura macquarii macquarii interleukin-36 receptor antagonist protein-like (LOC1371478...	Emydura macq...	83.6	83.6	90%	2e-17	33.33%	787	XM_067540265.1
PREDICTED: Trachemys scripta elegans interleukin-36 receptor antagonist protein-like (LOC117871587...	Trachemys scri...	82.4	82.4	85%	2e-16	34.84%	1434	XM_034759398.1
PREDICTED: Pelodiscus sinensis interleukin-1 family member 10-like (LOC142820928), transcript varia...	Pelodiscus sine...	79.0	79.0	82%	4e-15	35.25%	1692	XM_075910708.1
PREDICTED: Pelodiscus sinensis interleukin-1 family member 10-like (LOC142820928), transcript varia...	Pelodiscus sine...	79.0	79.0	79%	5e-15	35.82%	1690	XM_075910709.1
PREDICTED: Pelodiscus sinensis interleukin-1 family member 10-like (LOC142820928), transcript varia...	Pelodiscus sine...	78.6	78.6	79%	6e-15	35.82%	1857	XM_075910706.1
PREDICTED: Mauremys mutica interleukin-36 receptor antagonist protein-like (LOC123365076), transcri...	Mauremys mutica	75.5	75.5	53%	4e-14	43.96%	949	XM_045007706.1
PREDICTED: Pelodiscus sinensis interleukin-36 receptor antagonist protein-like (LOC102448715), trans...	Pelodiscus sine...	75.1	75.1	90%	9e-14	31.37%	1468	XM_075910776.1
PREDICTED: Carettochelys insculpta interleukin-36 receptor antagonist protein-like (LOC142004315), m...	Carettochelys in...	72.4	72.4	85%	2e-13	32.88%	653	XM_074981886.1
PREDICTED: Pelodiscus sinensis interleukin-36 receptor antagonist protein-like (LOC102448715), trans...	Pelodiscus sine...	73.6	73.6	90%	3e-13	31.37%	1428	XM_075910775.1
PREDICTED: Pelodiscus sinensis interleukin-36 receptor antagonist protein-like (LOC142820938), trans...	Pelodiscus sine...	71.6	71.6	88%	1e-12	31.13%	1290	XM_075910855.1
PREDICTED: Pelodiscus sinensis interleukin-36 receptor antagonist protein-like (LOC142820938), trans...	Pelodiscus sine...	71.2	71.2	88%	2e-12	31.13%	1290	XM_075910852.1
PREDICTED: Pelodiscus sinensis interleukin-36 receptor antagonist protein-like (LOC142820938), trans...	Pelodiscus sine...	71.2	71.2	88%	2e-12	31.13%	1180	XM_075910854.1
PREDICTED: Pelodiscus sinensis interleukin-36 receptor antagonist protein-like (LOC142820938), trans...	Pelodiscus sine...	70.9	70.9	88%	2e-12	31.13%	1283	XM_075910851.1

Figure 2: BLAST RESULTS

Download GenBank Graphics

Trionyx triunguis genome assembly, chromosome: 30

Sequence ID: OZ346534.1 Length: 17913073 Number of Matches: 1

Range 1: 1420032 to 1420238 GenBank Graphics ▼ Next Match ▲ Previous Match

Score	Expect	Method	Identities	Positives	Gaps	Frame
50.1 bits(118)	5e-05	Compositional matrix adjust.	26/69(38%)	41/69(59%)	1/69(1%)	+3

```
Query  91       VGEQPTLQLKEQKIMDLYGQPEPVKPFLFYRAKTGRTSTLESVAFPDWFIASSKRD-QPI  149
                +   P    ++++IMDLY   +  K F F    TG T   ESVAFP WF+++S +D QP+
Sbjct  1420032  LSHAPPPHPQDKEIMDLYQNSQEAKRFTFTCMATGSTRRFESVAFPGWFLSTSPKDGQPL  1420211

Query  150      ILTSELGKS  158
                 +T+  G++
Sbjct  1420212  RVTNISGEA  1420238
```

Figure 3: Alignment Details

Trionyx triunguis genome assembly, chromosome: 30 Sequence ID: OZ346534.1Length: 17913073Number of Matches: 1 Range 1: 1420032 to 1420238GenBankGraphicsNext MatchPrevious Match Alignment statistics for match #1 Score Expect Method Identities Positives Gaps Frame 50.1 bits(118) 5e-05 Compositional matrix adjust. 26/69(38%) 41/69(59%) 1/69(1%) +3 Query 91 VGEQPTLQLKEQKIMDLYGQPEPVKPFLFYRAKTGRTSTLESVAFPDWFIASSKR QPI 149 + P ++++IMDLY + K F F TG T ESVAFP WF+++S +D QP+ Sbjct 1420032 LSHAPPPHPQDKEIMDLYQNSQEAKRFTFTCMATGSTRRFESVAFPGWFLSTSP-KDGQPL 1420211

Query 150 ILTSELGKS 158 +T+ G++ Sbjct 1420212 RVTNISGEA 1420238